Advances in CMOS Technologies and Future Directions for Cell Height scaling

Anupama Bowonder

Foundry Technology Development

Intel Corporation

Outline

- Artificial Intelligence and Computational Requirements

- Advanced CMOS Innovations
  - Gate All Around Transistors
  - Backside Power Delivery

- Enhanced cell scaling technologies
  - Fork Sheet Transistors
  - Complimentary FETs

- Summary

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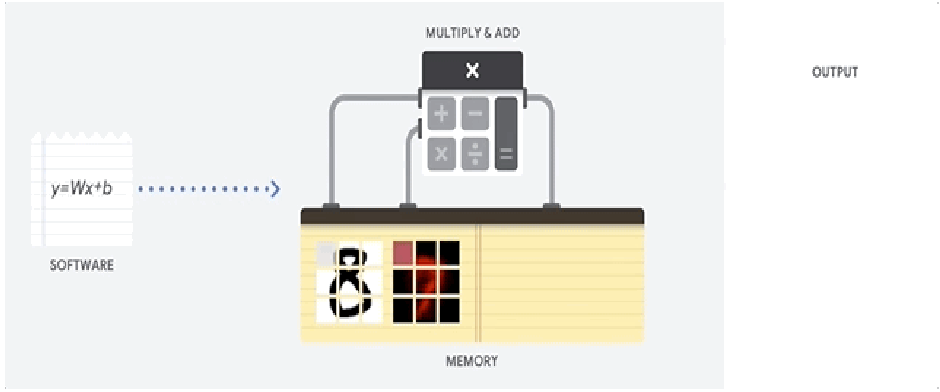
Evolution from CPUs to GPUs

CPUs : Flexibility



- ❑ Load many kinds of software for many different types of applications. (Word processing, control rocket engines, execute bank transactions, classify images with a neural network)
- ❑ CPU loads values from memory, performs a calculation and stores the result back in memory for every calculation.
- ❑ Memory access is slow when compared to the calculation speed and can limit the total throughput of CPUs.

GPUs : Higher throughput

- ❑ GPUS work for applications with massive parallelism. GPU can provide an order of magnitude higher throughput than a CPU.
- ❑ 1000s of logic units in a single processor to execute thousands of multiplications and additions in parallel.
- ❑ GPU still general-purpose and has to support many different applications and software.
- ❑ For every calculation in the thousands of ALUs, a GPU must access registers or shared memory to read operands and store the intermediate calculation results.

Evolution from GPUs to AI Chips

GPUs

- Processor for graphics applications, graphics rendering and processing.
- GPU chips have faster speeds and higher parallel processing in graphics rendering.
- Used for scientific computing, cryptography, machine learning. **Focus on graphics rendering.**

Common GPU chips:

NVIDIA's GeForce, Quadro, Tesla
AMD's Radeon, FirePro
Intel's Xe

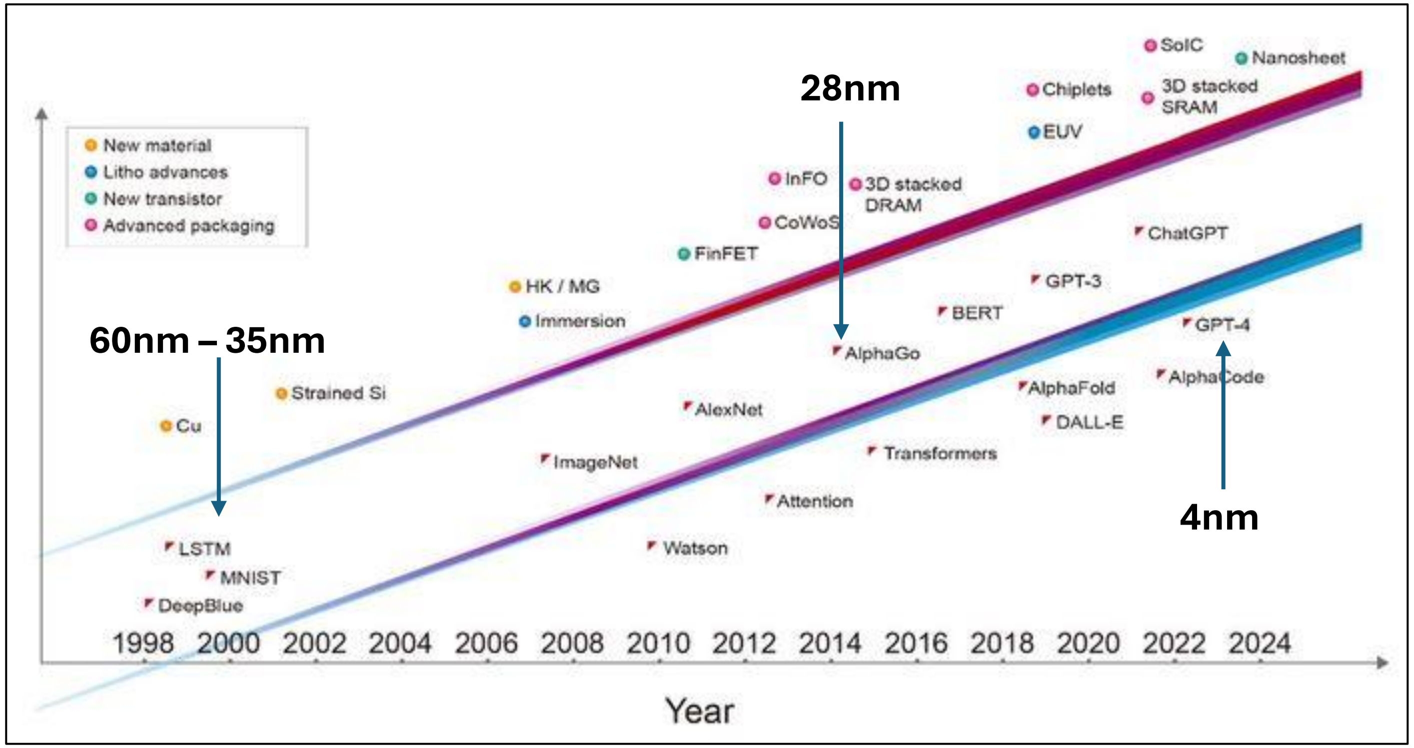
AI Chips

- Processor to support the computing needs of artificial intelligence applications.
- Usually use highly optimized algorithms and can quickly handle large numbers of matrix calculations, vector calculations, neural network calculations and other tasks.
- Used in computer vision, speech recognition, natural language processing, machine learning. **Focus on supporting neural network calculations**.

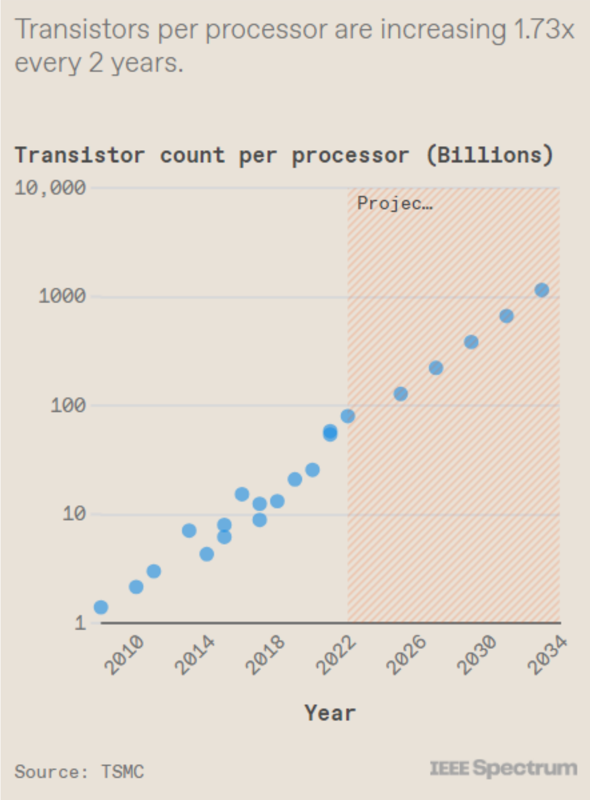
Common AI chips:

NVIDIA’s Tesla V100, A100, T4
Google's TPU (Tensor Processing Unit)

Technology scaling: Backbone of Artificial Intelligence



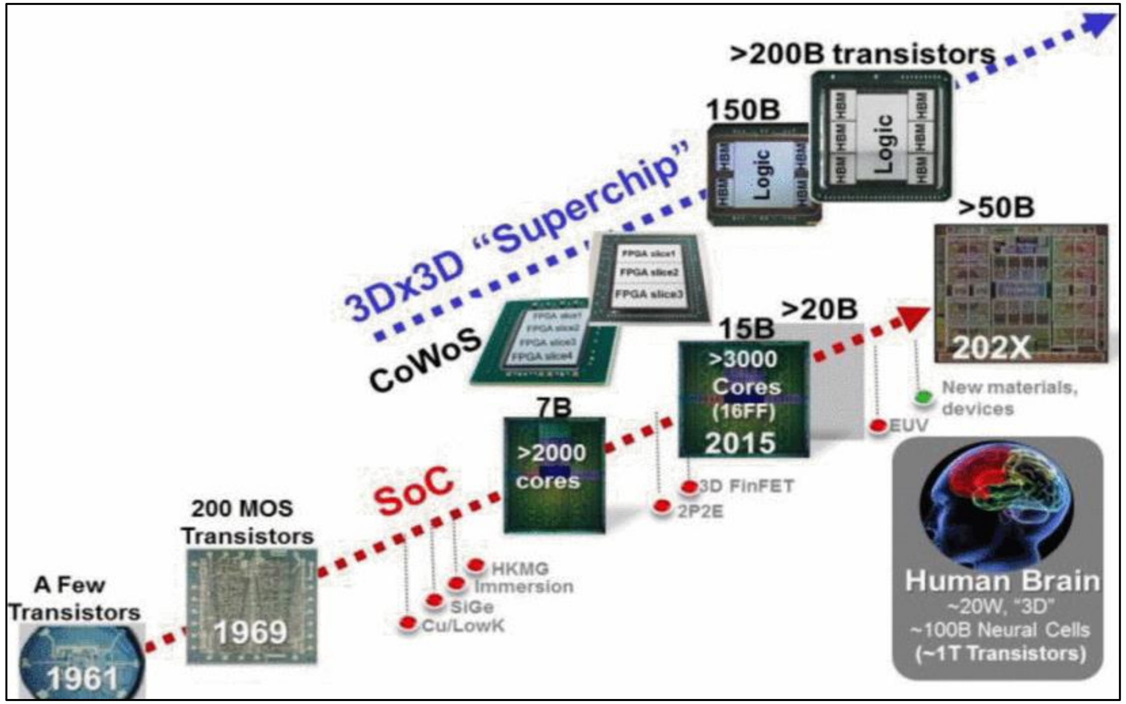
IEEE spectrum, March 2024



IEEE spectrum, March 2024

- ❑ For the AI revolution to continue, will need semiconductor innovation.
- ❑ 2D Scaling has transitioned into 3D Scaling
- ❑ Within the decade need for a 1 Trillion transistor GPU .

Increased Computation: 3D Stacking



- ❑ Increased computation + Energy-efficient 3Dx3D system scaling needed
- ❑ 3Dx3D system scaling: 3D CMOS scaling + 3D hetero system integration technologies, (CoWoS®, InFO).
- ❑ Transistor count multiplies beyond traditional CMOS SOC with more stacked chips + package level interconnect.

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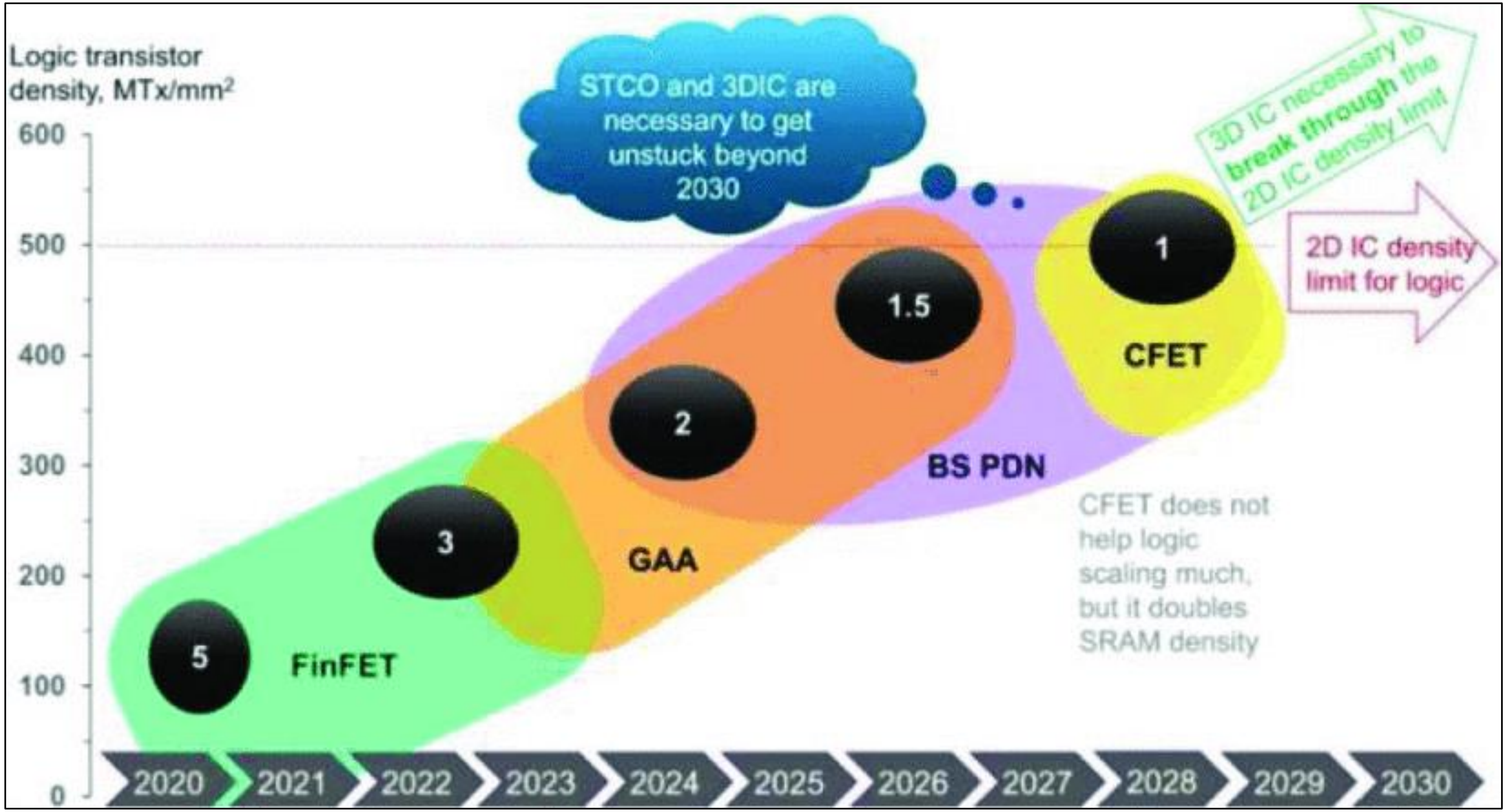
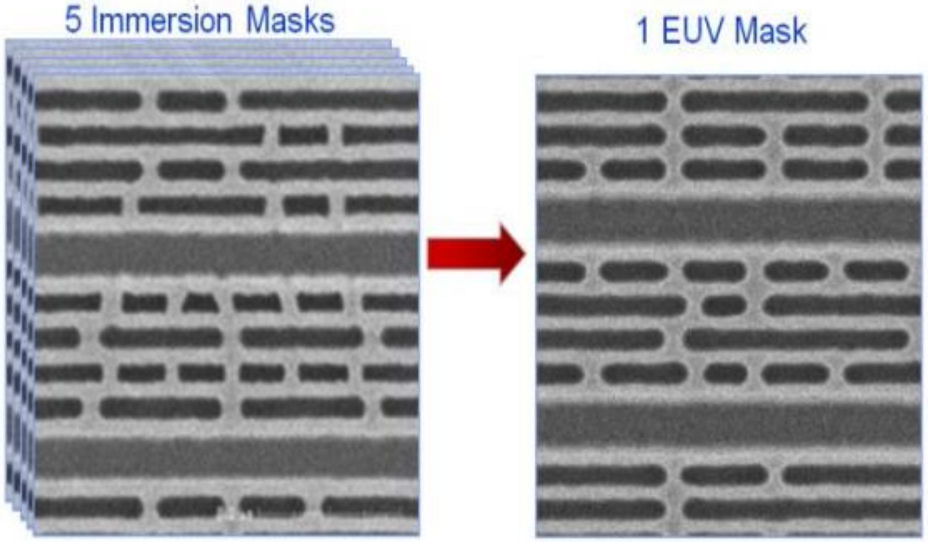
Logic Technology Roadmap/Horizon By Synopsis

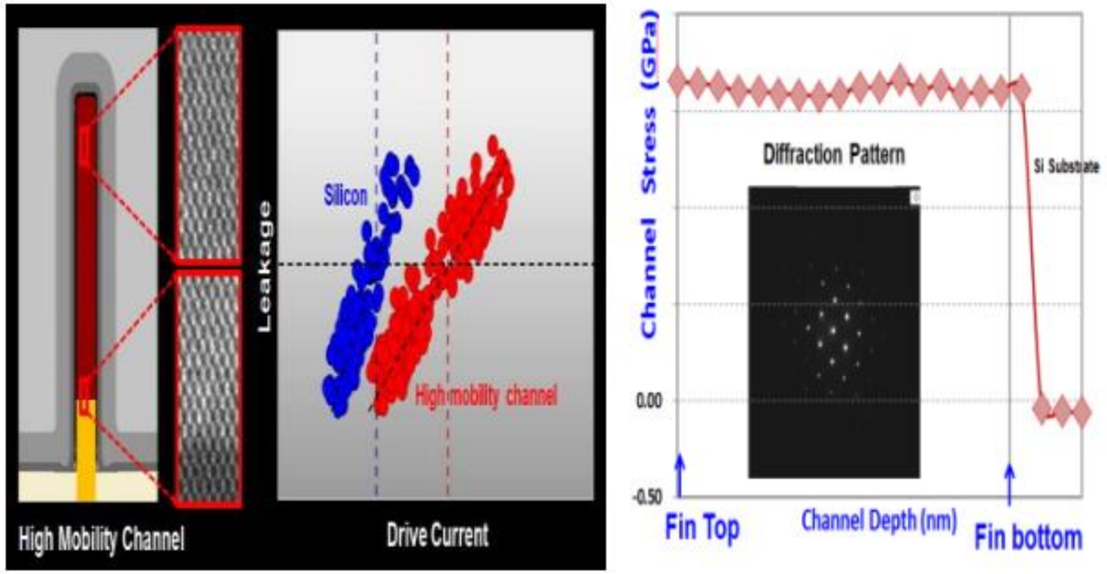


Semiconductor Engineering, April 2023

Industry First High Mobility Channel Introduction

2019 IEDM, TSMC



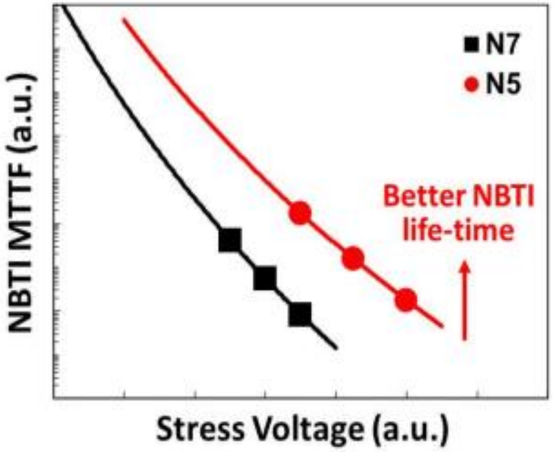


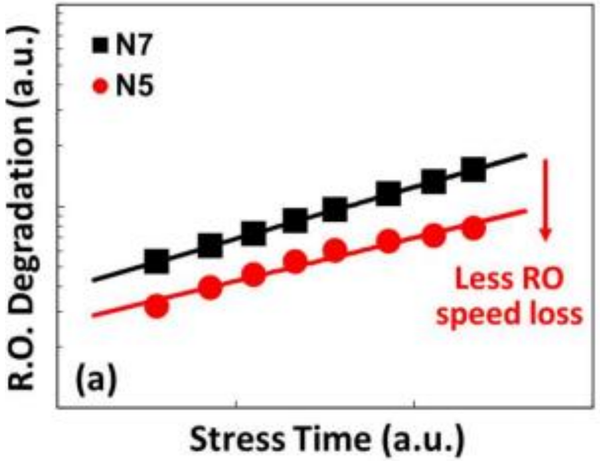
1 EUV mask replacing 5 Immersion Masks. Better patterning fidelity, shorter cycle time, less defects

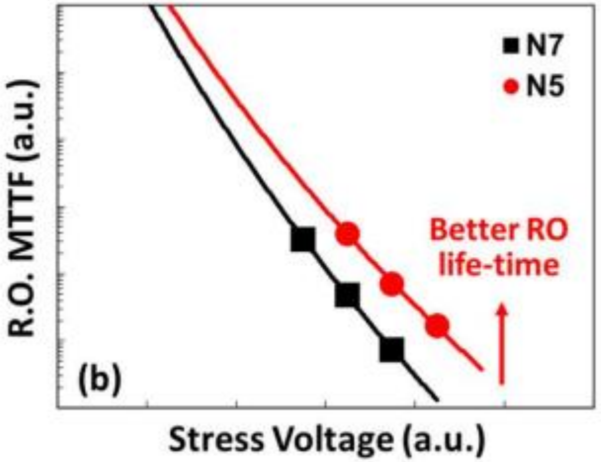
18% performance gain with High mobility channel vs Si. TEM showing fully strained HMC lattice constant with Si Lattice constant

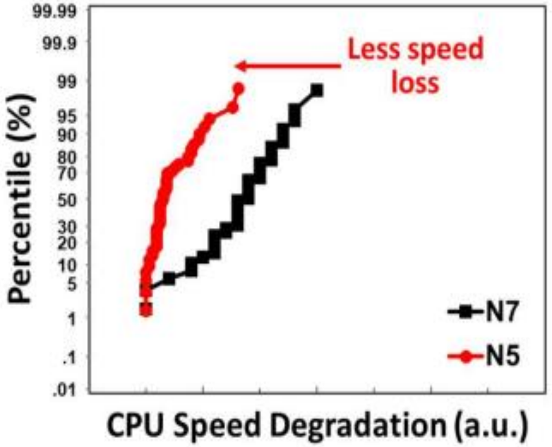
FinFET: Improved Reliability

2019 IEDM, TSMC









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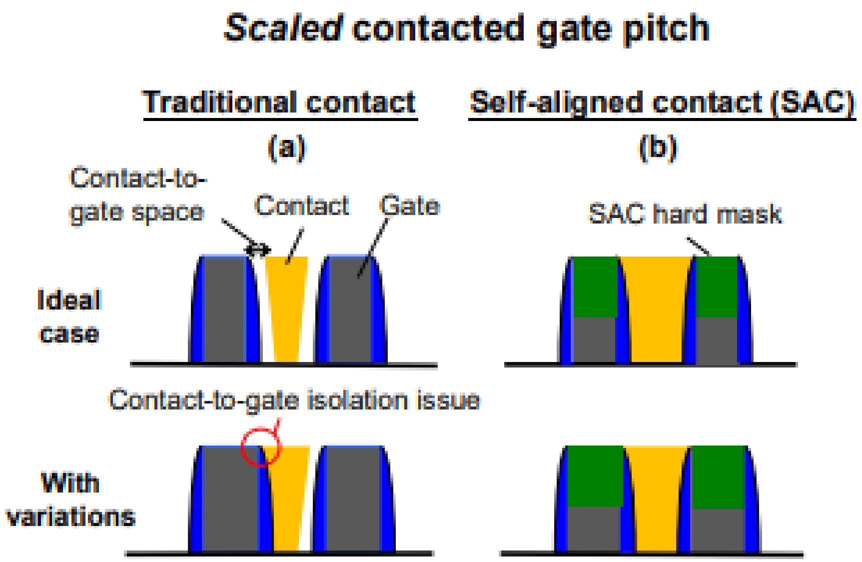
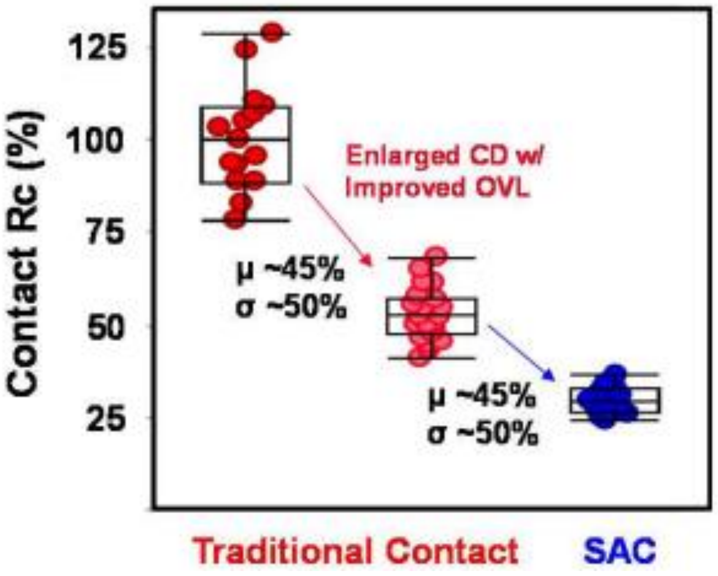
FinFET Process Developments: Self Aligned Contacts

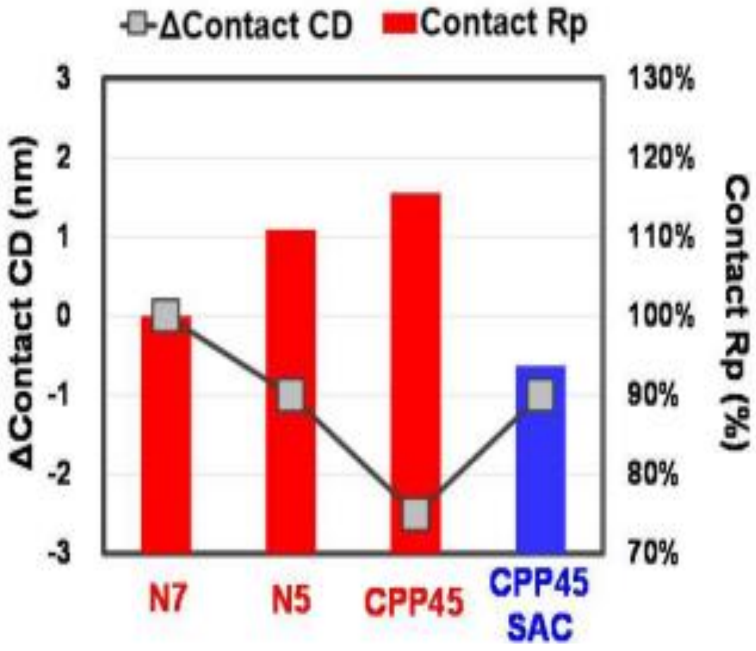
2022 IEDM, TSMC



Self aligned Contacts not prone to variations induced contact-to-gate isolation issue



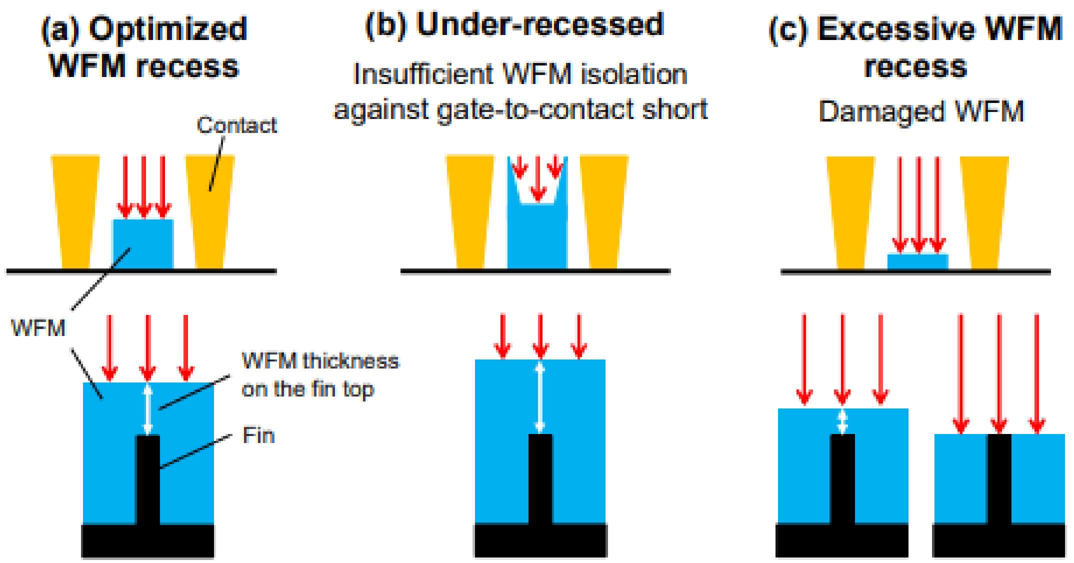
SAC reduces resistance by 45% and variation by 50%



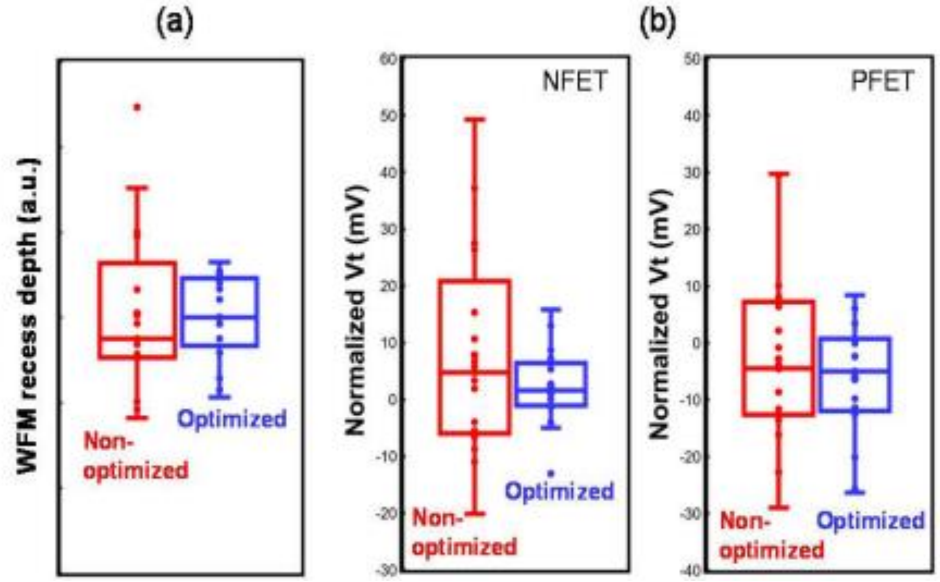
SAC disrupts CD reduction while providing even smaller resistance

FinFET: Process Developments: Vt Uniformity

2022 IEDM, TSMC



Self Aligned Contact flow WFM and precise depth control



Optimized SAC WFM recess process with uniform depth cross wafer (a) significantly reduces the Vt variations (b)

FinFET: Process Development Cell Height Scaling

2022 VLSI

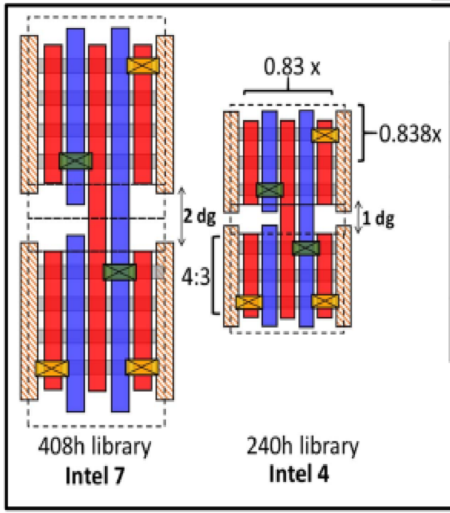
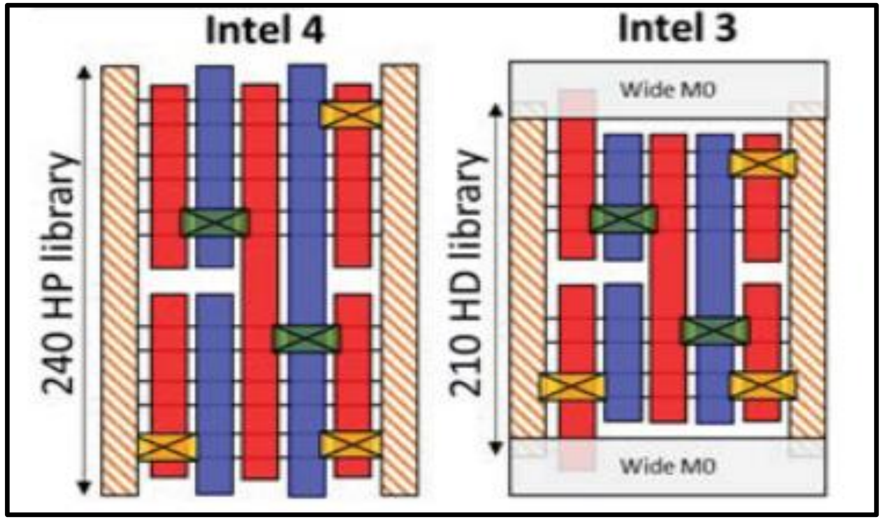
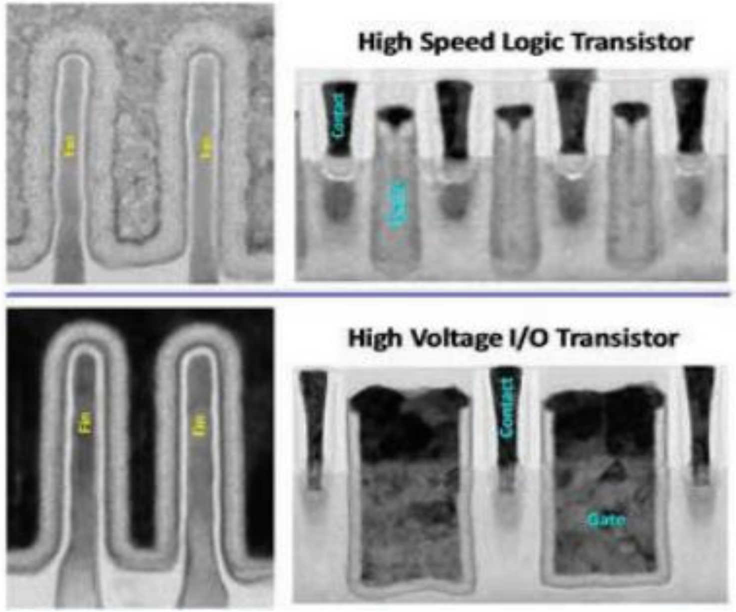


Table 1	Intel 7	Intel 4	Scale
Contacted Gate Pitch (nm)	54/60	50	0.83x
fin Pitch (nm)	34	30	0.88x
Interconnect Pitch (nm)	40	30	0.75x
Library height	272/ 408	240	

2023 VLSI



2023 VLSI

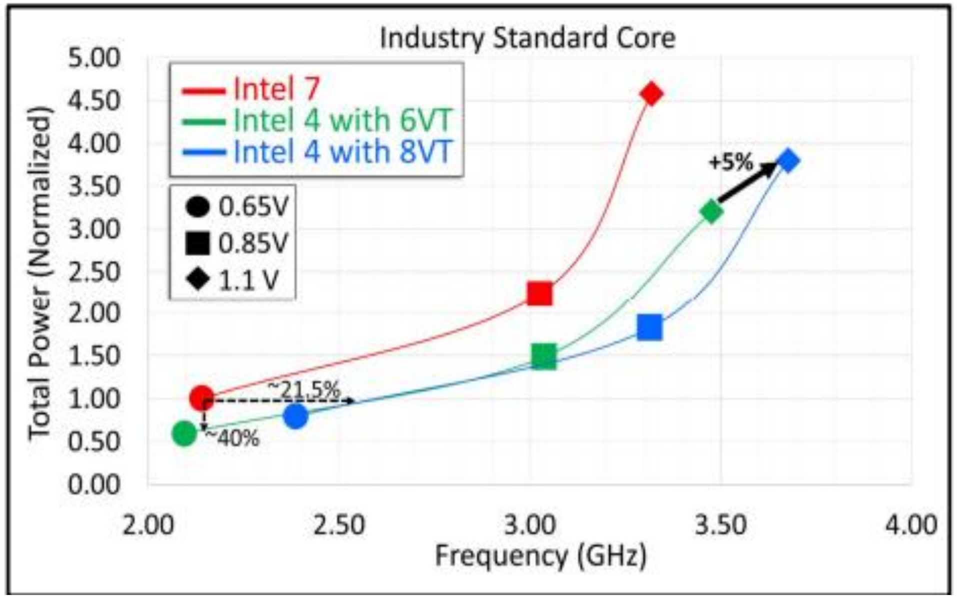


Core Logic Library scaled fom Intel 7 to Intel 4 to Intel 3

- Continual Cell height scaling and intrinsic transistor SCE improvement.
- Intel 3, 5x reduction in offstate leakage at matched drive currents to Inel 4
- Optimizations in contact resistance and gate to contact capacitance. Intel 3 15% higher Frequency than Intel 4 at matched leakage

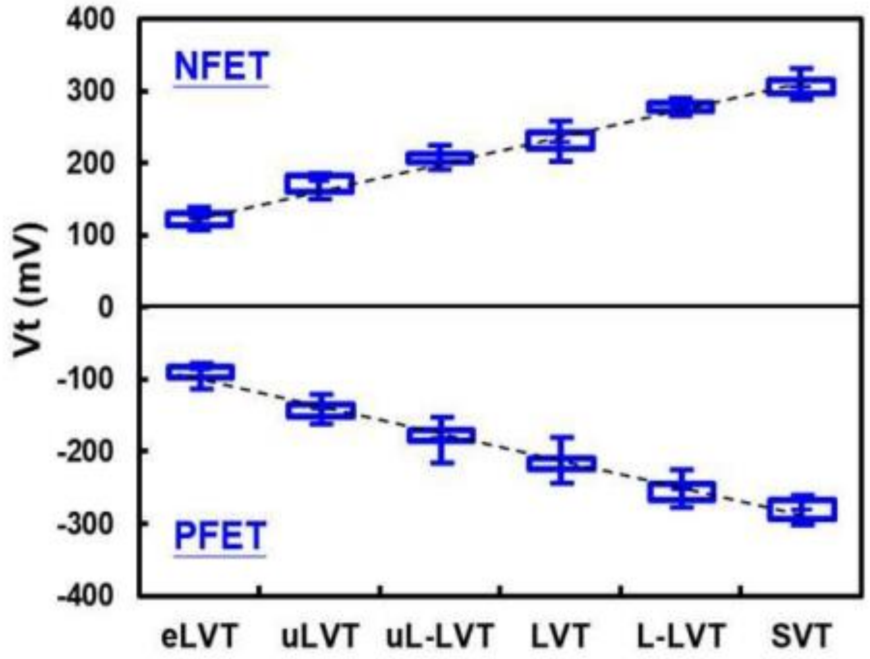
FinFET: Multi Vt Process Needs

2022 VLSI, Intel



Circuit analysis of core shows increased Performance at matched power. 8Vt flow enables 5% performance over 6Vt at high Vcc

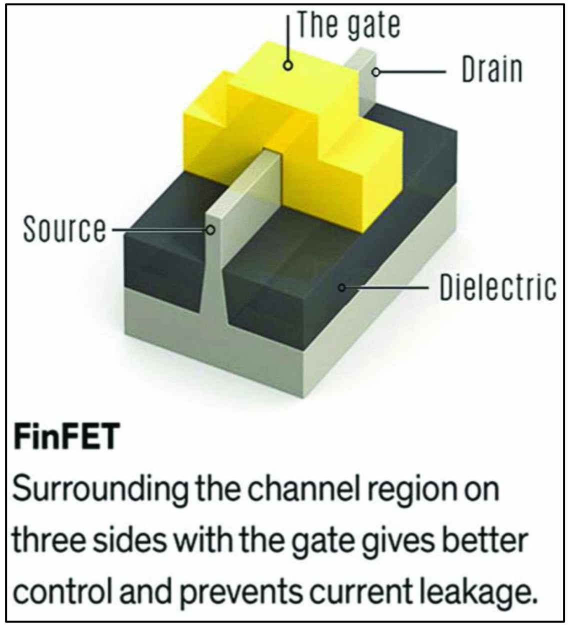
2023 VLSI, TSMC

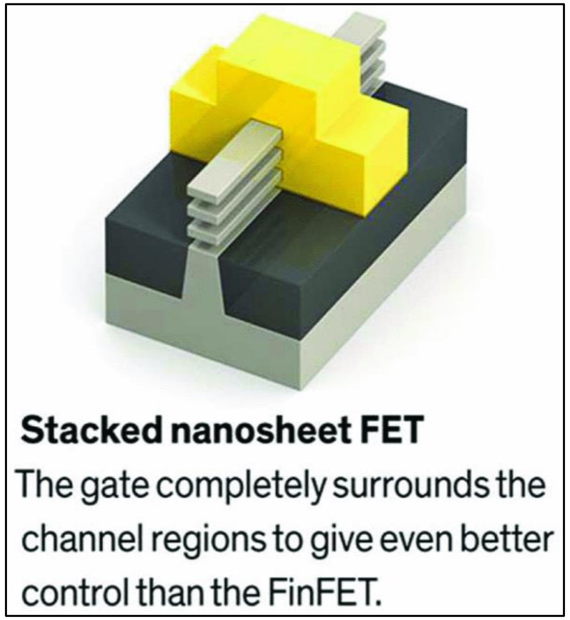


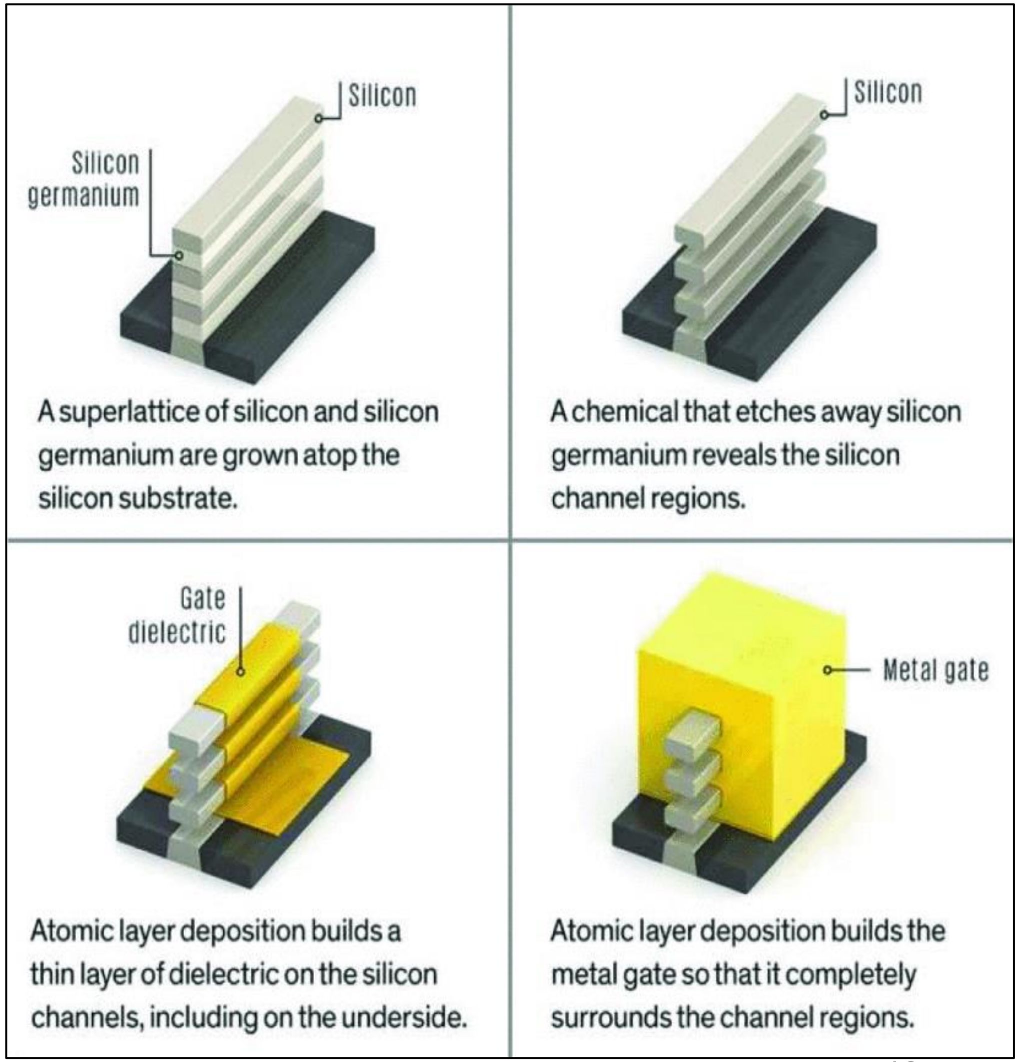
TSMC, multi Vt offering spanning 200mV

Finfet optimizations involving multi Vt offerings with reduced local layout effects

Advanced CMOS Transistors: Gate All Around Transistors

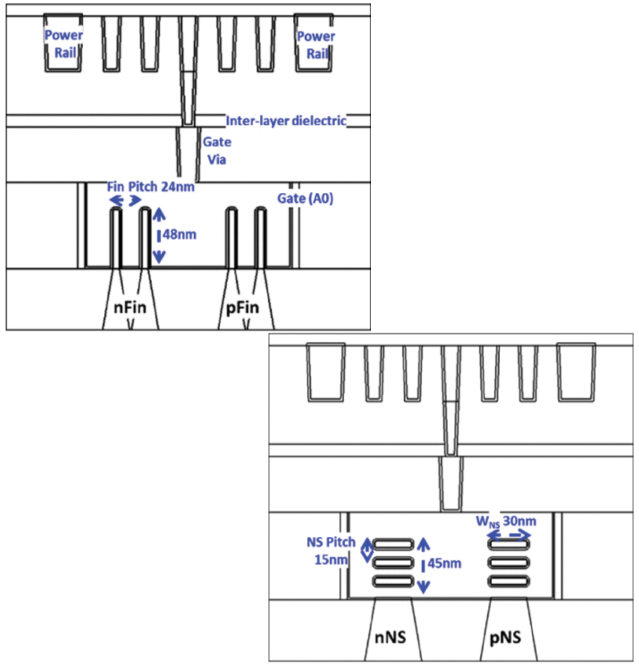


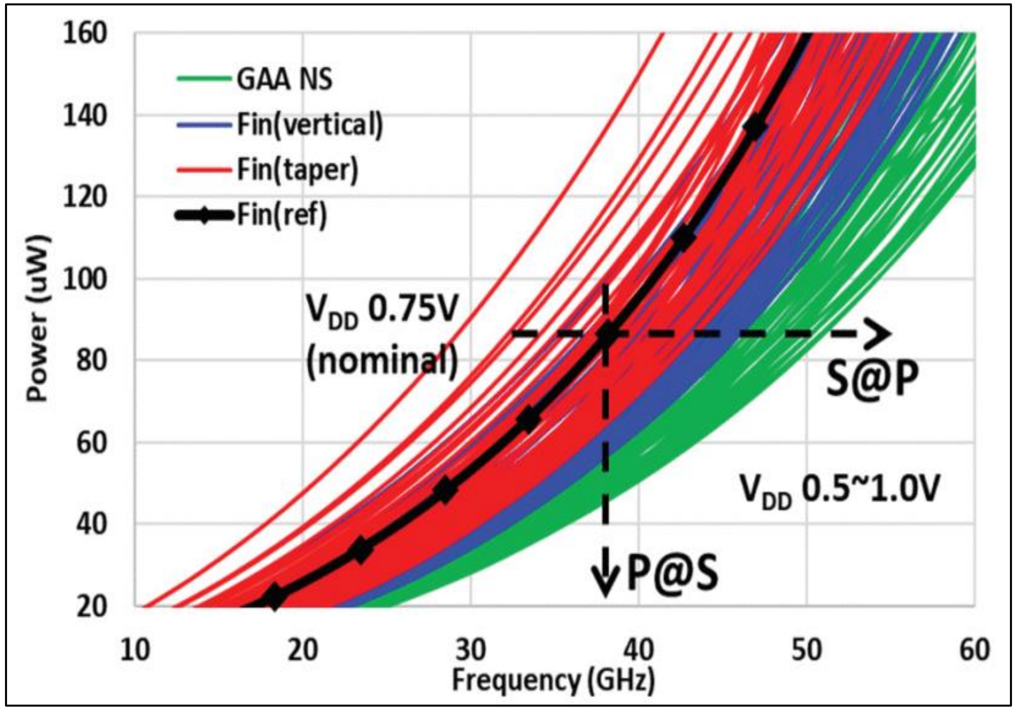


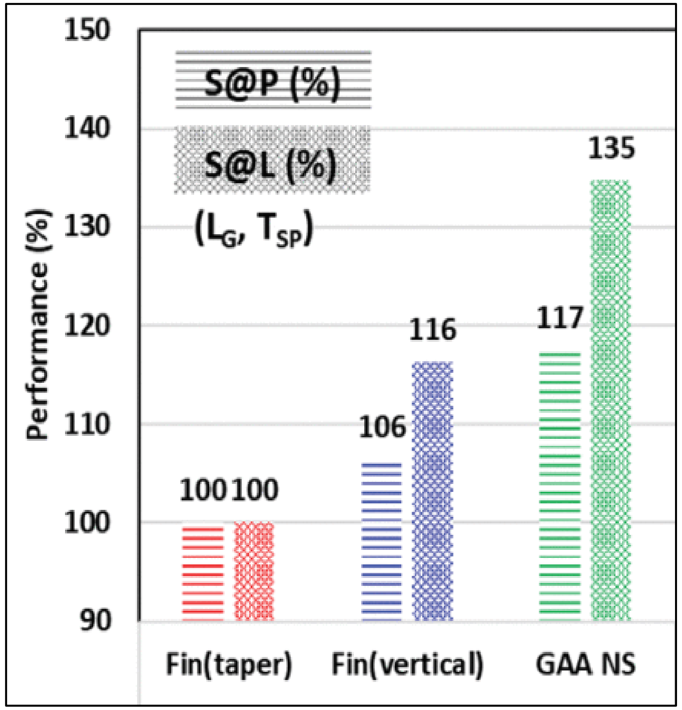


Current Industry Focus:
- TSMC – Gate-All-Around FET (GAAFET)
- Samsung – Multi-Bridge Channel FET (MBCFET)
- Intel – NanoRibbon transistor

Gate All Around Transistors: Power Performance Area (PPA) evaluations

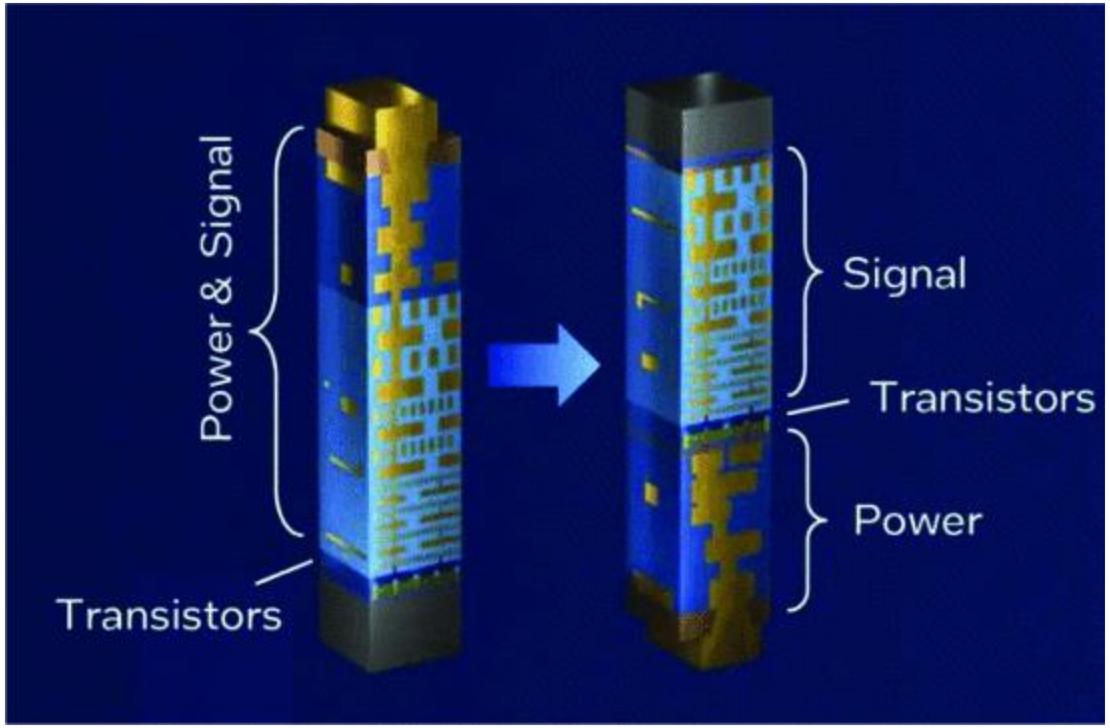


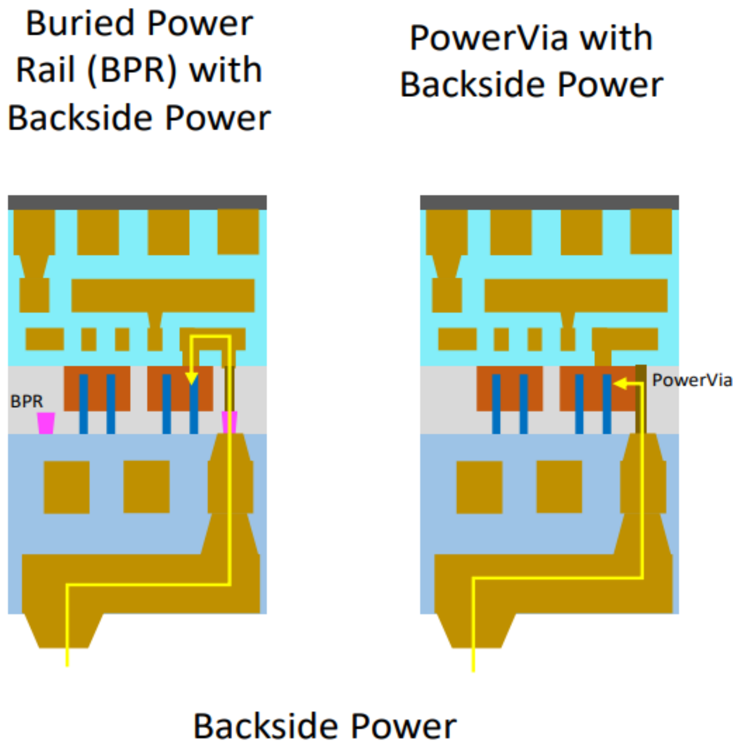




- **Fin/NS devices speed-at-iso-power and speed-at-iso-leakage (0.75 V)**
- **Fin (vertical) provides 6%/16% speed-at-iso-power and speed-at-iso-leakage gain, while GAA NS improves both by 17%/35%.**

Back Side Power Delivery Network (BS PDN)

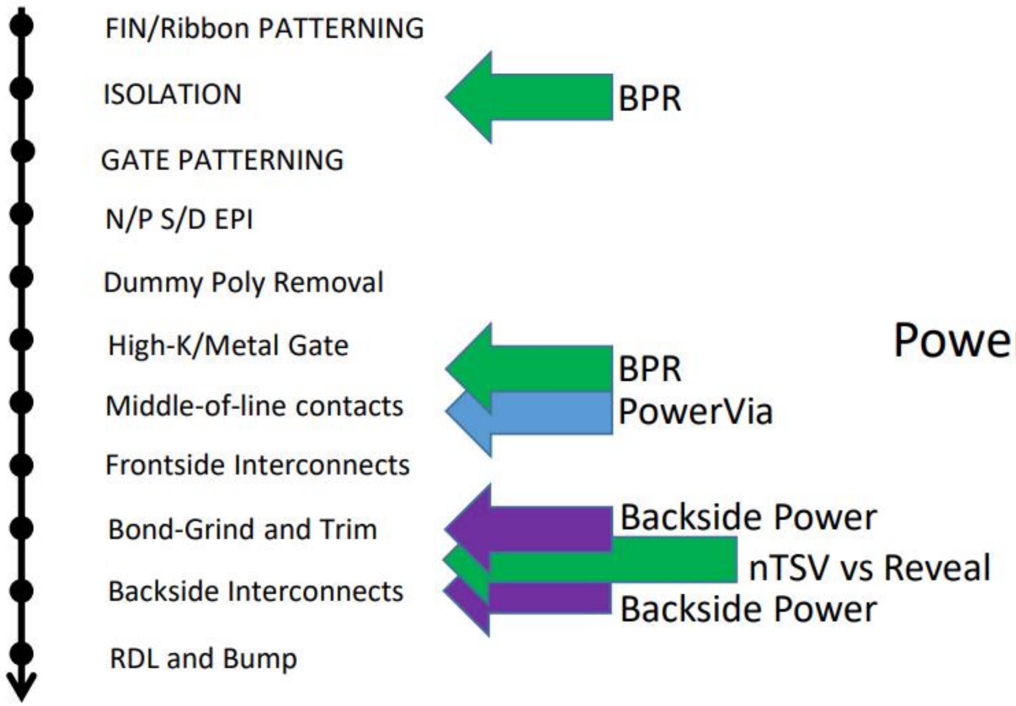




When signal interconnects and power rails are above the transistors challenges for scaling.

Relocate the power bus, below the transistor layer, leave signal interconnections above transistor layer.

Back Side Power: Architecture Options



nTSV

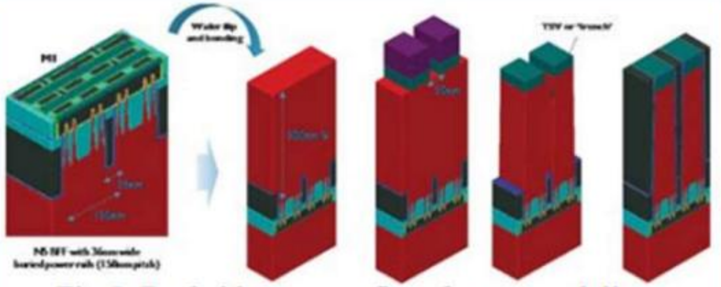
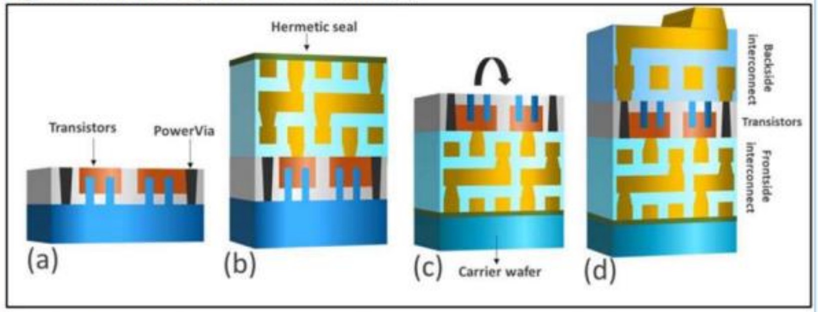


Fig 5: Backside process flow for power delivery

J. Ryckaert et al., EDTM 2019

PowerVia

Fig. 4. Schematic of process flow details. (a)



finFET patterning + PowerVia integration with S/D contacts. (b) BE interconnects optimized for signal routing. (c) Carrier bond → wafer thinning → PowerVia reveal. (d) BE interconnects optimized for power delivery.

Backside power Architecture options: nTSV and PowerVia

- BPR introduces early front end and mid section process modules
- PowerVia limits process module adds to midsection Once connection is revealed there is little difference in process

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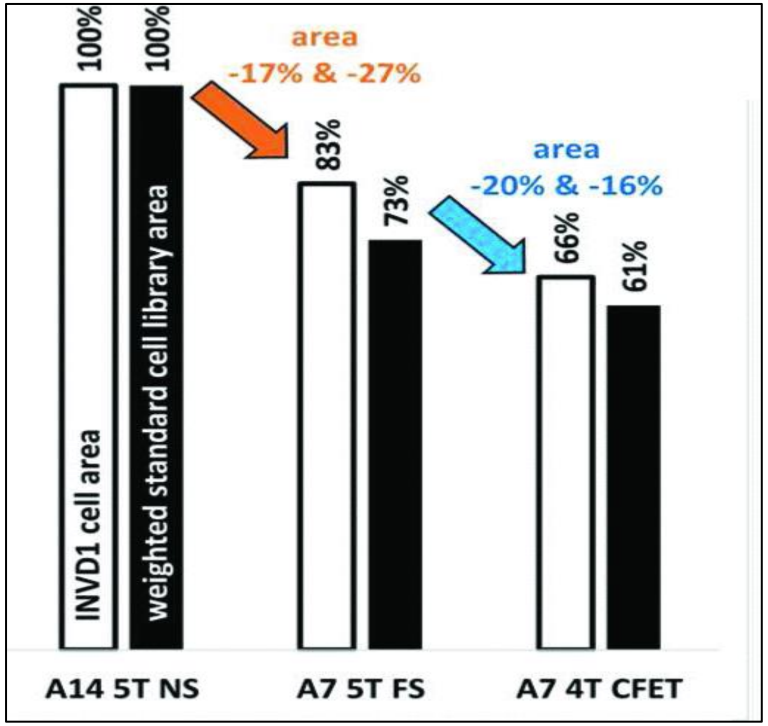
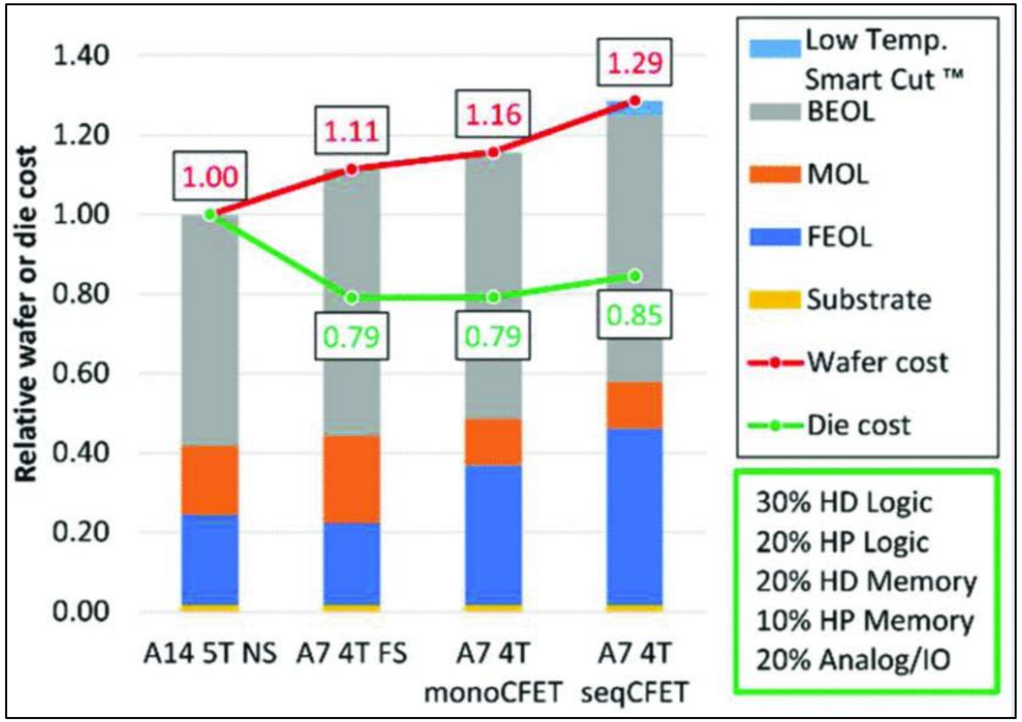
Advanced cell height scaling technologies: Fork Sheet and CFETs



Area scaling A14 5T down to A7 4T CFET cell & library level (~Arm 64bit core PnR).



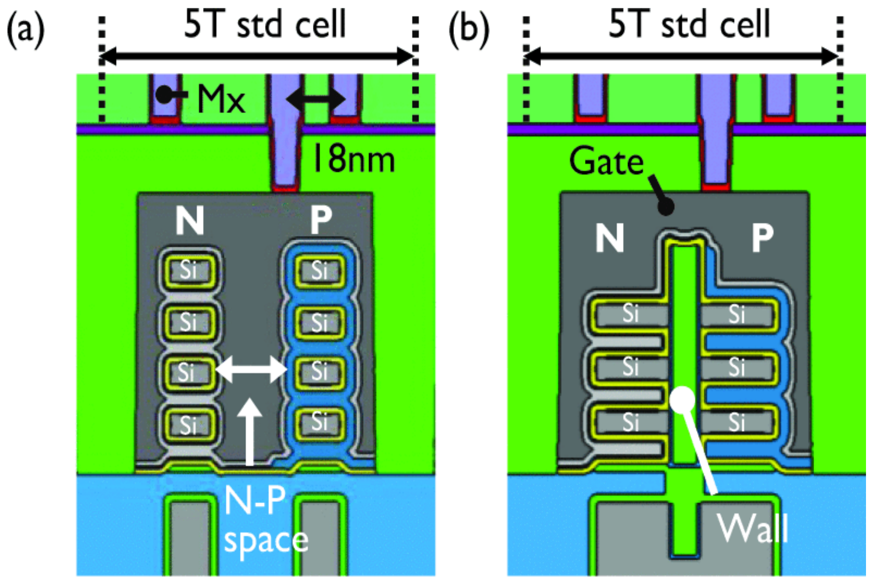
Wafer & Die cost of A7 4T new tech vs A14 5T NS.

Advanced technologies Wafer Cost ($$$) vs Die Cost!

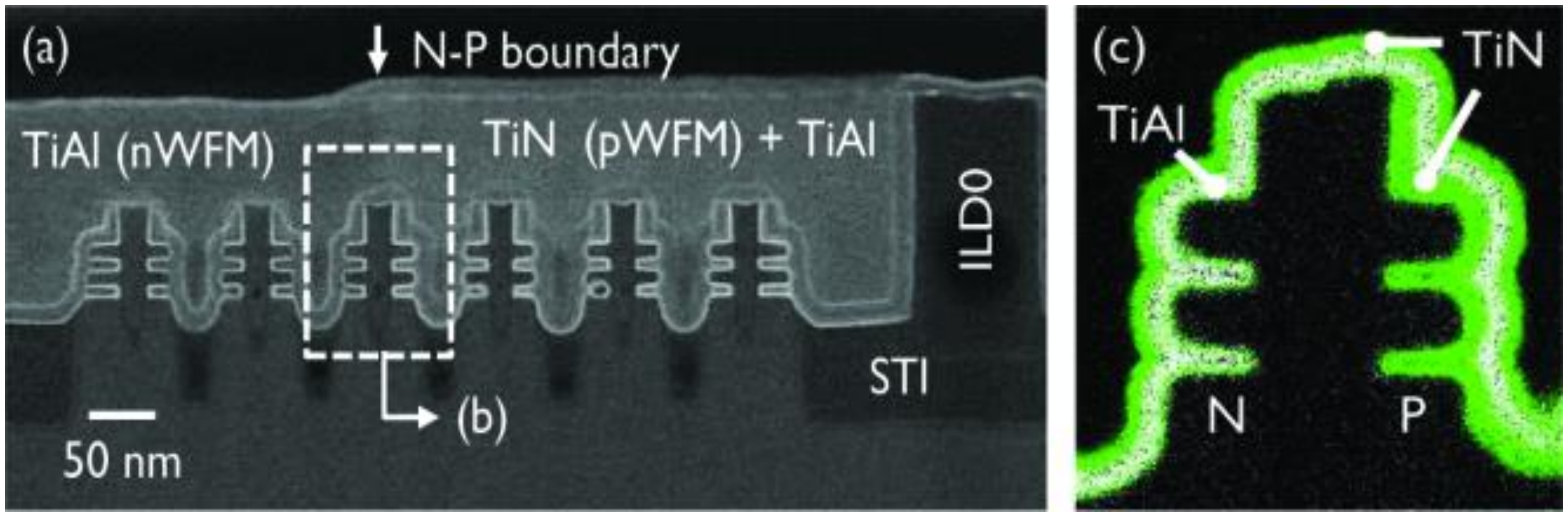
- Adv cell height scaling options (FS and CFETs) increase wafer cost.
- Fork sheet wafer cost from added metal layers and CFETs form additional process steps.
- CFETs superior area scaling enables better die cost than FS.

Advances CMOS for Cell Height scaling: Fork Sheet Transistors

VLSI 2021



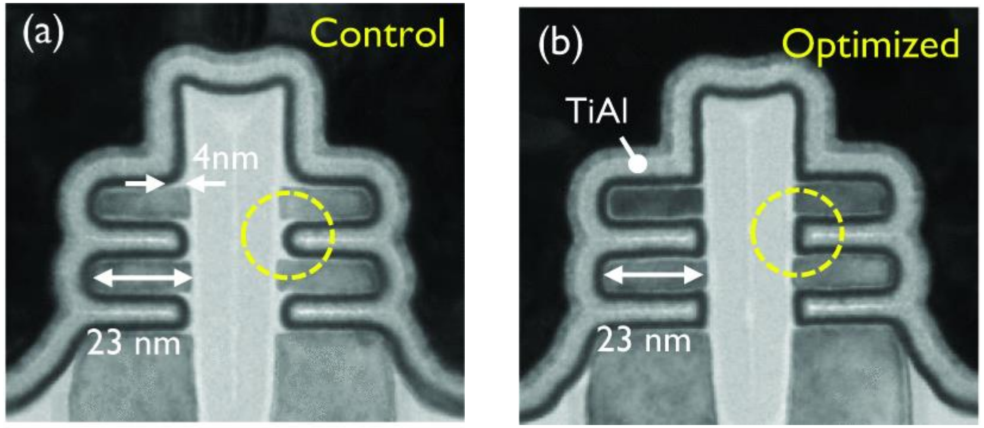
VLSI 2021



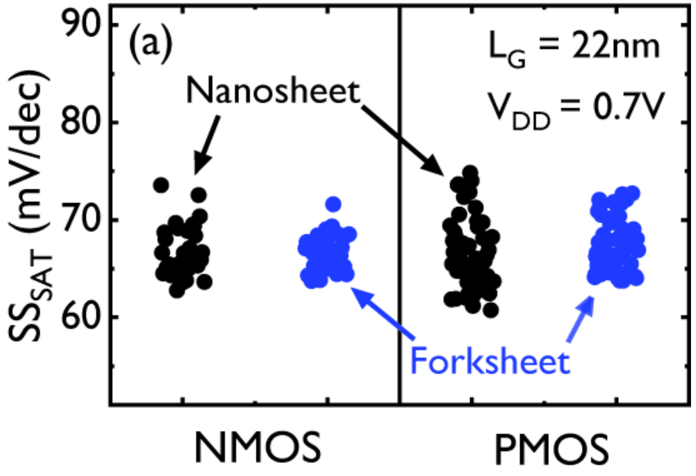
- Forksheet and nanosheet structures on the same wafer with and without sub-20nm fin-to-fin spacings.
- SiGe/Si fins self-aligned double patterning. SiN wall by conformal deposition and isotropic etchback
- Forksheet dual work function metal with the N-P boundary on top of the forksheet wall.
- Different WFM layers on either side of a 17nm wall (N-P space) TiN and TiAl elemental mapping

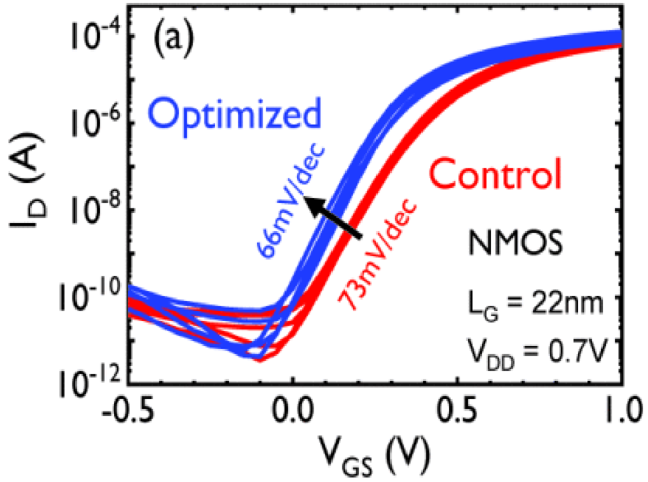
Advances CMOS for Cell Height scaling: Fork Sheet Transistors

VLSI 2021



VLSI 2021

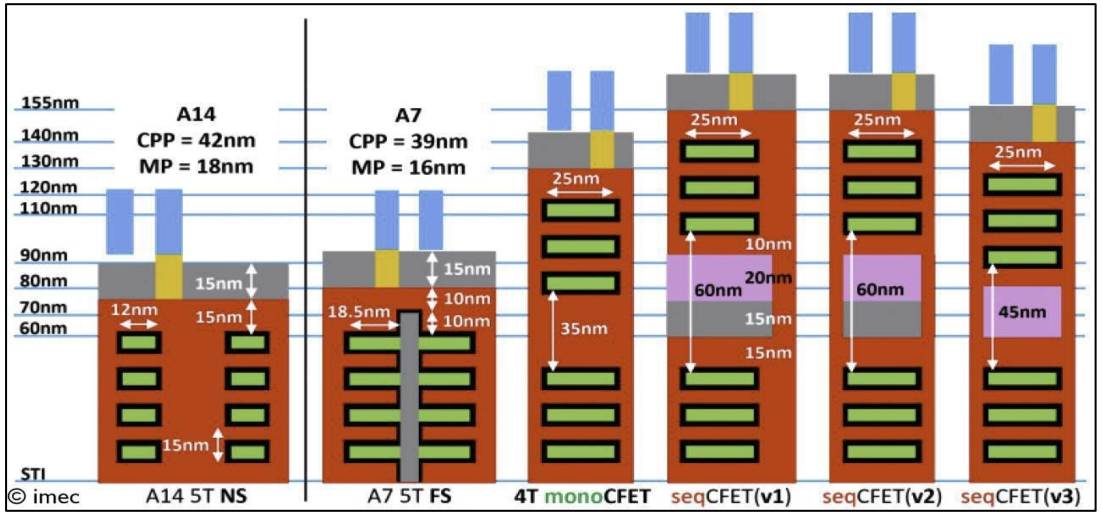




- Tri Gate geometry can cause SS_{SAT} degradation of 1-4 mV/dec at 16nm $L_{G.}$
- Improved gate/channel overlap shows improved drive current enhancement

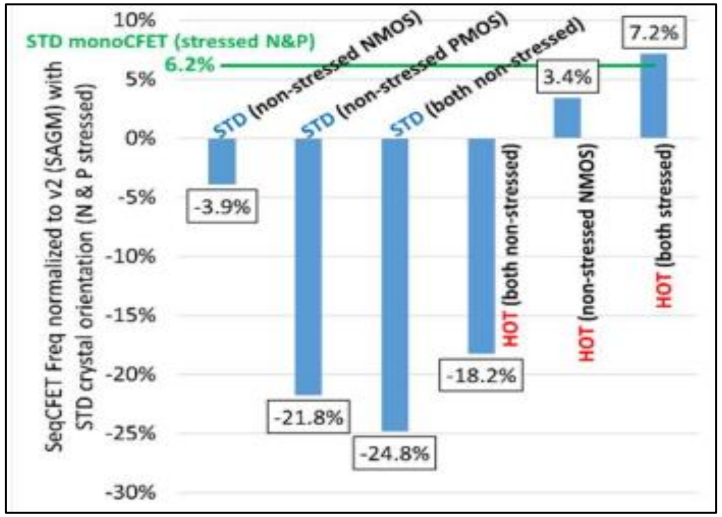
CFETs: Area and Performance: Monolithic vs Sequential Integration

VLSI 2022



Nansoheet (NS), forksheet (FS) and CFET (monolithic and sequential).

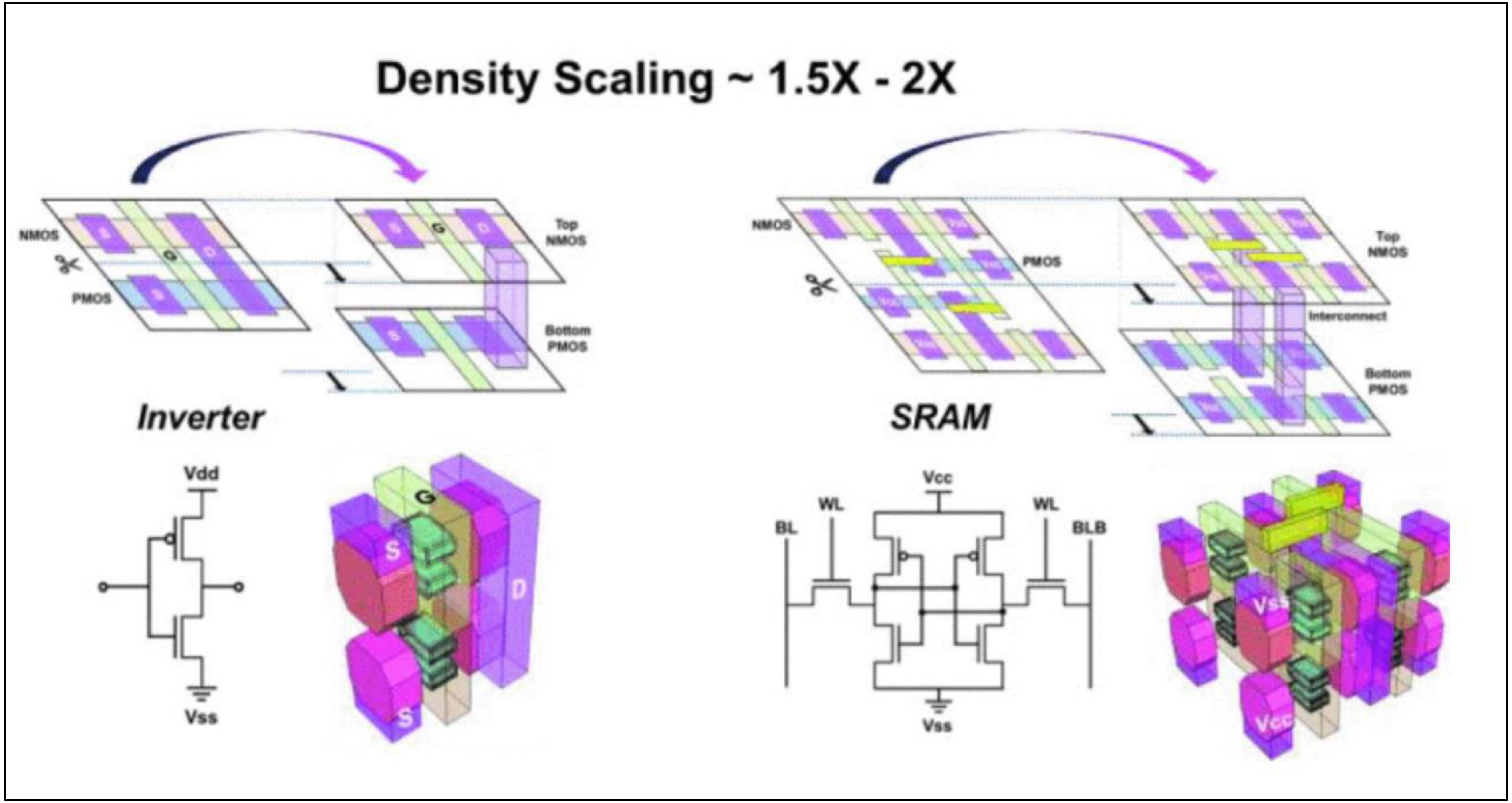
VLSI 2022



Hybrid Orientation (HOT) boosts frequency by 7%

- CFETs monolithic process flow potentially consume less area, outperform their sequential counterparts, with high capacitance.
- Critical sequential CFET optimizations needed to optimize area and performance:
  (1) self-aligned gate merge ((v2) in the figure)
  (2) omission of the gate cap (v3)
  (3) Use hybrid orientation technology. N on P config: Si <100> wafer top for high N mobility + Si <110> bottom wfr for high P mobility

CFETs: Area Scaling Benefits



VLSI 2023

CFETs: Advanced processing for cell height scaling

Lead process flow approaches for CFET

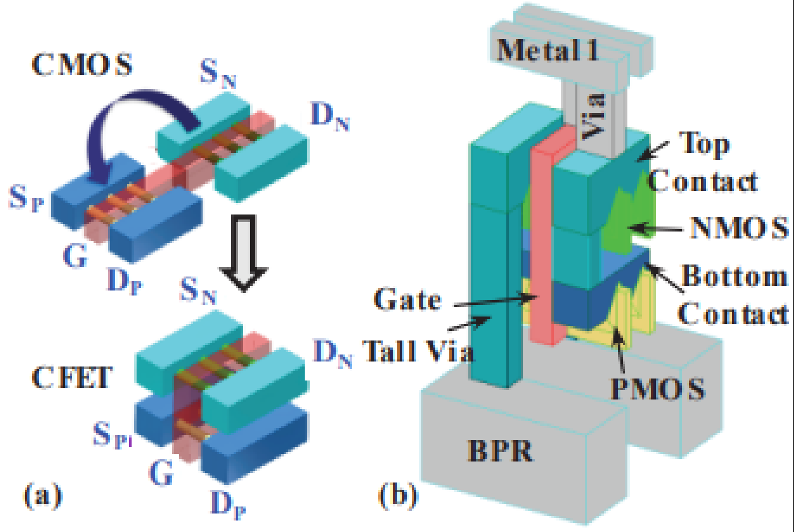


Fig. 1. (a) Folding an NMOS on top of a PMOS results in a CFET. (b) 3D architecture of the CFET device.

<u>Monilithic Integ</u>

- Stacked on single substrate
- Top Bottom self alignment
- Least disruptive vs Nano sheet flow

Challenges:

- High Aspect Ratio patterning
- Selective deposition and removal of material
- Deposition of high quality Epi Films

<u>Sequential Integ</u>

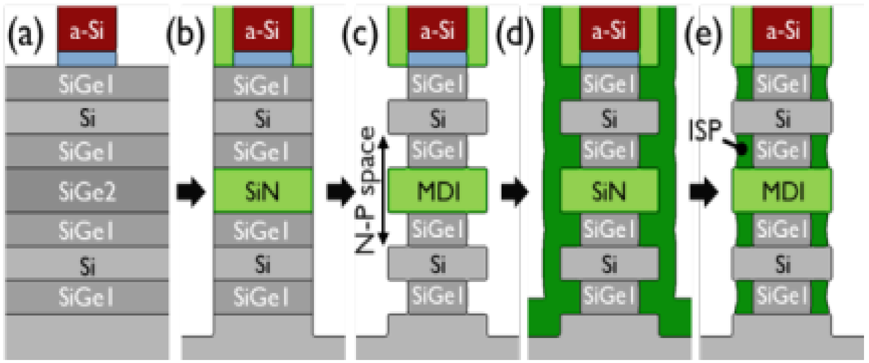
- FEOL matched to non CFET
- Layer Transfer and thinning of the bonding dielectric (20nm) between tier

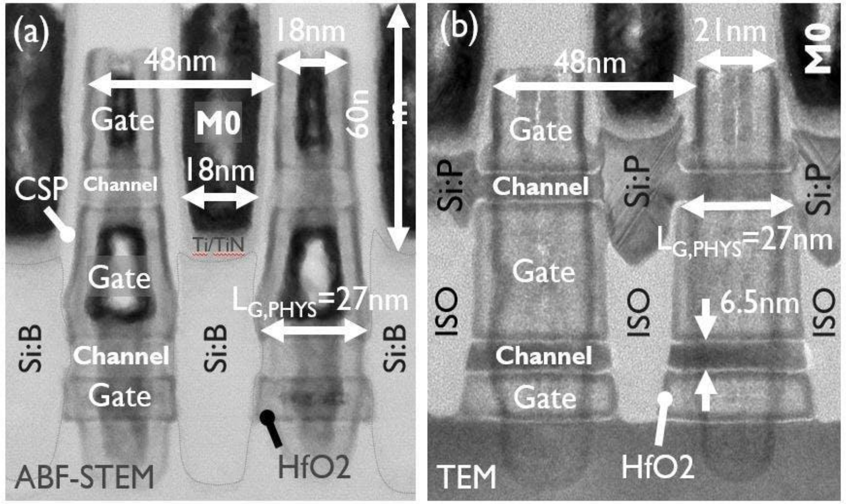
Challenges:

- Shrinking standard cells needs CFETs in combination with advanced technology to contact the stacked transistors + advanced wafer transfer

CFETs: Monolithic Integration

VLSI 2023, IMEC





End-of-process (a) bottom pFET and (b) top nFET (LG,PHYS=27nm)

IEDM 2023, TSMC

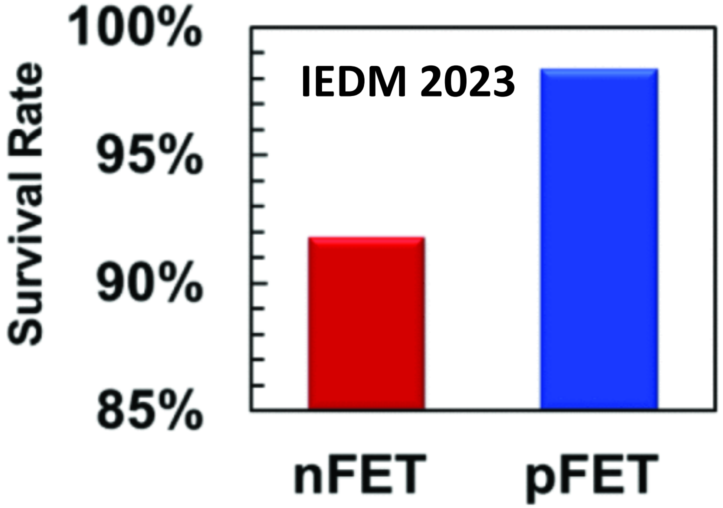
Scaled CFET	This work	VLSI 2023 [14]	IEDM 2021 [13]
Organization	TSMC	IMEC	Intel
Device Architecture	Monolithic CFET	Monolithic CFET	Monolithic CFET
Top Device	1-Nanosheet Si nFET	1-Nanosheet Si nFET	2-Nanoribbon Si nFET
Bottom Device	1-Nanosheet Si pFET	1-Nanosheet Si pFET	2-Nanoribbon Si pFET
Gate Pitch	48 nm	48 nm	55 nm
Gate Length	15 nm	27 nm	19 nm
Inner Spacer	Yes	No	?
N/P SD Epitaxy Stacking	Yes	No	Yes
N/P SD Isolation	Yes	No	No
nFET SS	75 mV/dec	70 mV/dec	70 mV/dec
pFET SS	73 mV/dec	73 mV/dec	N/A
	@ V_{ds} = ± 0.75V	@ V_{ds} = ± 0.7V	@ V_{ds} = 0.65V
I_{OFF}(nFET): I_{OFF}(pFET)*	~ 1:1	~ 1:100 – 1:1000	N/A
nFET I_{ON}*	> 1 mA/μm	~ 0.4 mA/μm	~ 0.2 mA/μm
pFET I_{ON}*	> 1 mA/μm	~ 0.1 mA/μm	N/A
Note: V_{gs_off} as V_g @ I_{ds} = 1 nA/μm	@ $V_{gs} - V_{gs\_off} = V_{ds}$ = ± 0.75V	@ $V_{gs} - V_{gs\_off} = V_{ds}$ = ± 0.7V	@ $V_{gs} - V_{gs\_off} = V_{ds}$ = 0.65V

Monolithic CFET: Epitaxial growth of the bottom channel, deposition of an intermediate sacrificial layer and last epitaxial growth of the top channel.

VLSI 2023: IMEC demo unipolar CFET devices through monolithic integration at 48nm gate pitch:

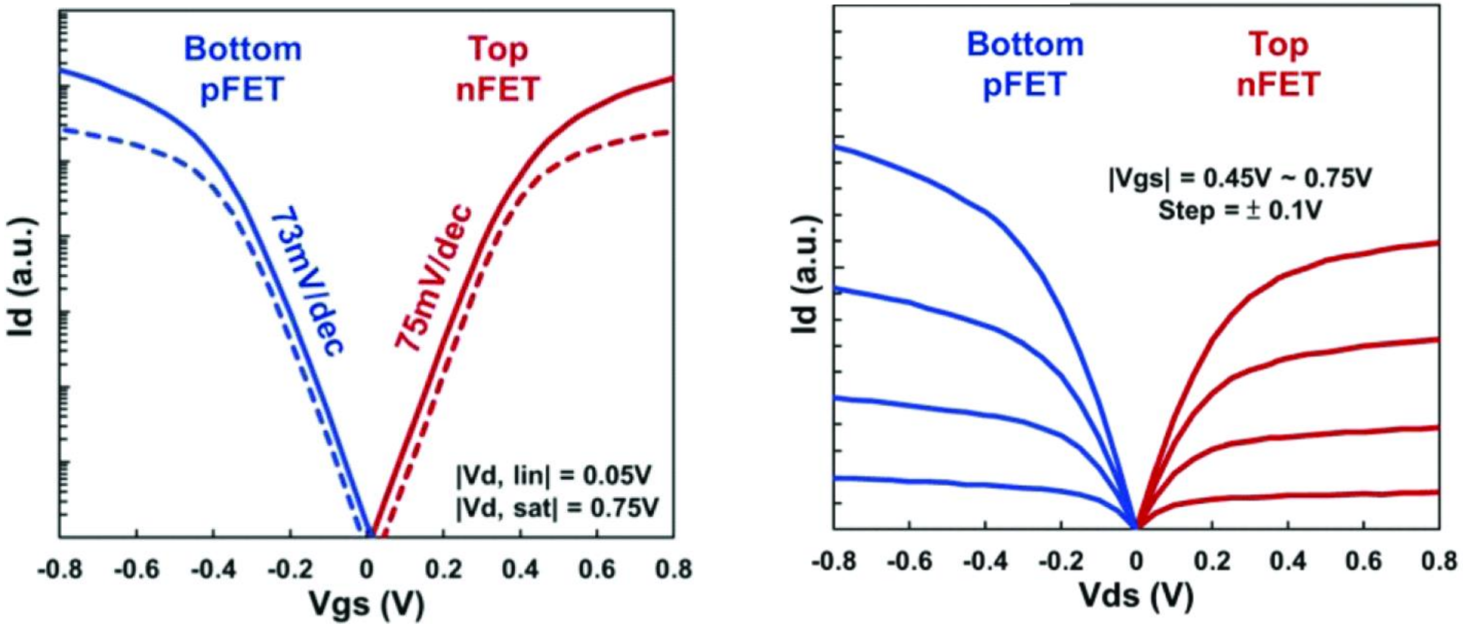
- The source-drain epitaxial growth and source-drain contacts are evaluated for bottom and top devices.
- Limited aspect ratio for first demonstration, only one nanosheet for the top and bottom device each.
- With a 30nm vertical separation between top and bottom sheets, we found a way to independently contact top and bottom devices.

CFETs: Device Characteristics



Parameter	Survival criterion
$V_{T,lin}$ (V), $V_{T,sat}$ (V)	$\|V_{T,lin}$ (V)$\|$<1 and $\|V_{T,sat}$ (V)$\|$<1
DIBL (mV/V)	10< DIBL (mV/V) <200
Subthreshold Swing (SS_{sat})	10 <SS_{sat} <100
Rch (Ω*um)	10< Rch (Ω*um) <350

TSMC late news IEDM 2023



PFET performance is slightly better than nFET.

Degraded nFET parasitic resistance. Optimizing the n/p SD ISO dielectric recess profile and nEPI deposition needed.

CFETs: New Process Challenges and Opportunities

Middle dielectric isolation (MDI):

- 1) Need to create a vertical dielectric isolation between top and bottom gate to differentiate the Vt setting between top and bottom devices.
- 2) Ge-rich sacrificial layers are converted into the MDI dielectric creating the n-p WF metal separation in gate.

Vertical Isolation between S/D contact metals of Top and Bottom Devices:

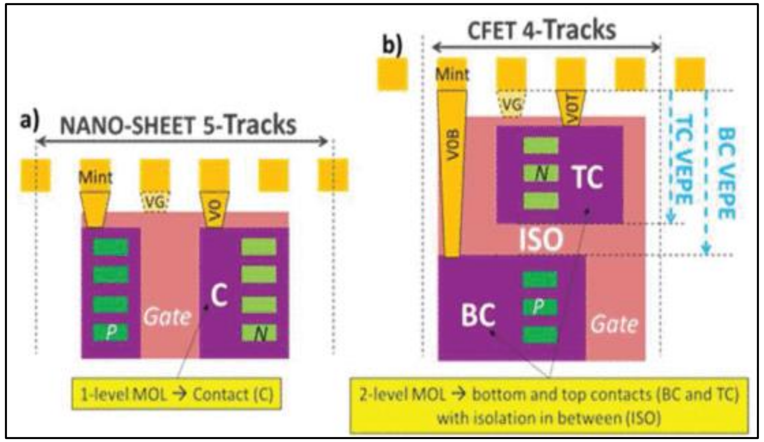
- Need to isolate bottom and top contacts between two tall gates and route bottom and top transistors

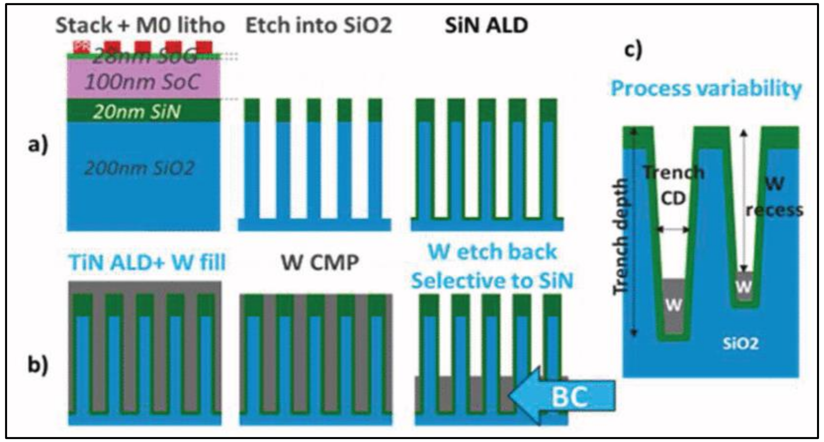
Growing differentially doped epi on bottom and top devices.

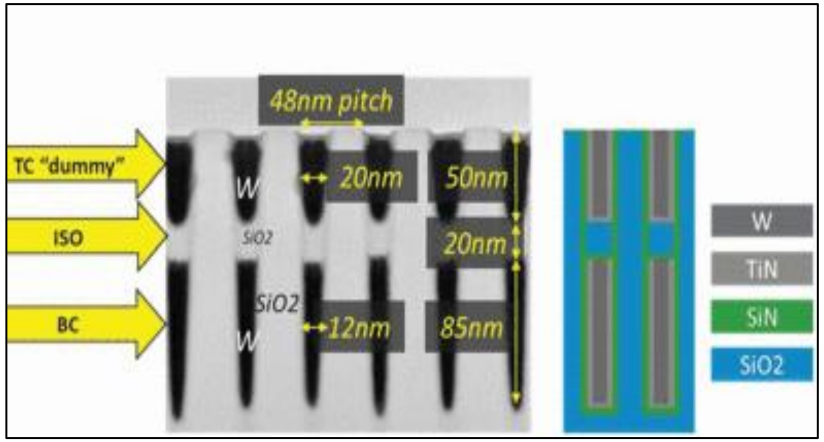
Backside power delivery: Connecting active devices from the backside

CFETs: Middle EOL process development

VLSI 2023







- **Bottom FET contacts (BC) need metal lines built deep in between tall gates.**
-
- **Need MDI/ISO in place before building the top contacts (TC) for connection to the top transistor.**
- **Short-loop flows to mimic tall gates and exercise high AR metallization and etch back.**

CFETs Sequential Integration: Challenges and Opportunities

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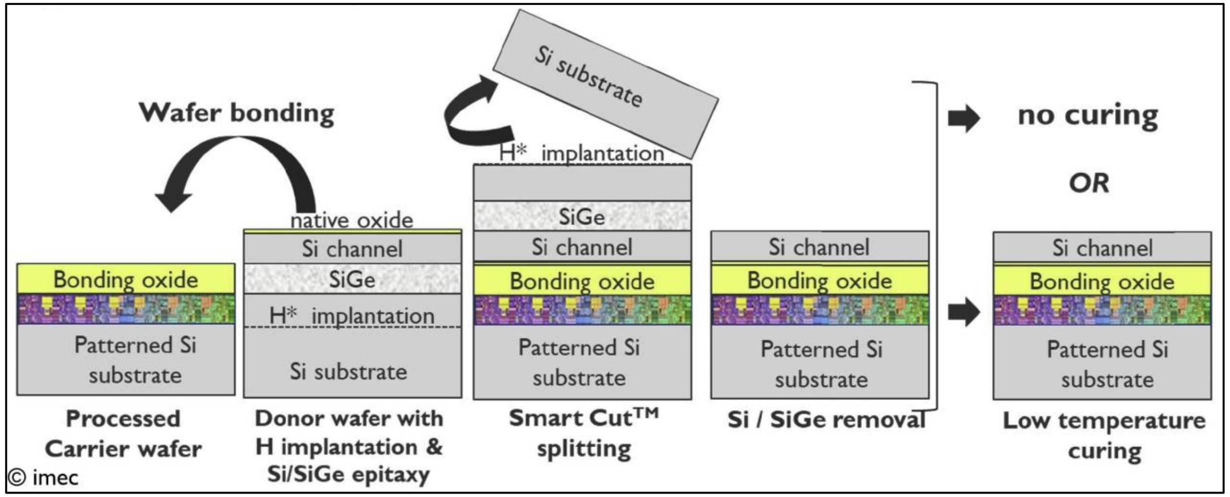
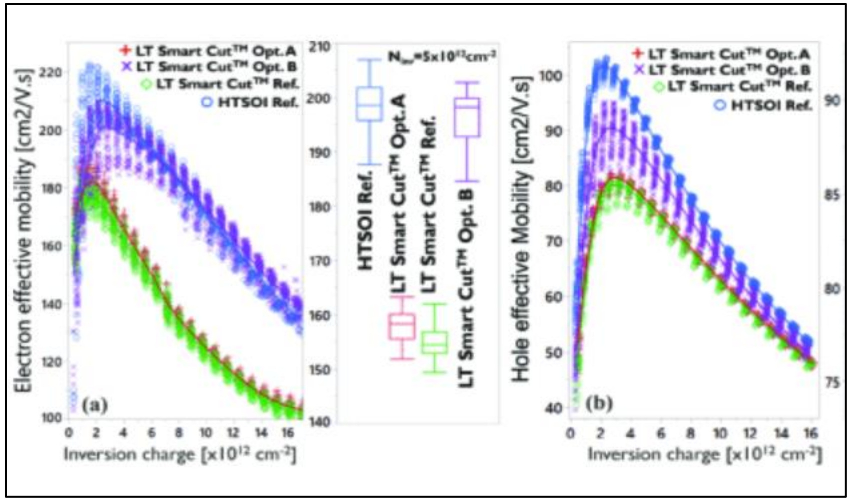


SOITEC low-temperature Smart Cut^{TM} layer transfer flow, with no curing or low-temperature curing

VLSI 2022



Effective field mobility vs. inversion charge. Improved mobility with additional low-temperature curing step

Lead Approaches for Layer Transfer:

- SOITEC's low-temperature Smart Cut^{TM} flow attractive for cost. Engineered bulk donor wafer enables thin layer splitting at low temperature. Donor wafer reusable ($$$ Saving).
- Substrate removal by grinding and Si etch-back,

Post smart cut optimization of electron mobility of top devices needed

Outline

- Artificial Intelligence and Computational Requirements
- Advanced CMOS Innovations
  - FinFETs and Gate All Around Transistors
  - Backside Power Delivery
- Enhanced cell scaling technologies
  - Fork Sheet Transistors
  - Complimentary FETs
- Summary

Summary

- CMOS scaling is alive and ever evolving into 3D space

- Increase computer power:

  3D CMOS Scaling

  +

  3D Hetero System integration technologies